

数学物理方程(理科版)

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July 22, 2020

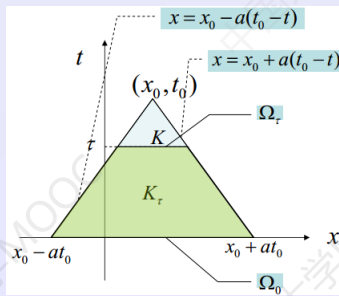
第二章 波动方程

§2. 初值问题（一维情形）

2.4 能量不等式

一、记号

如图：设 (x_0, t_0) 为平面直角坐标系 xot 上半平面上任一点，过此点向下作二条特征线 $x = x_0 \pm a(t_0 - t)$ ，这二条特征线与 x 轴围成一个三角形区域，称此区域为以 (x_0, t_0) 点为顶点的特征锥，记之为 K 。



二、能量不等式

定理2.3 设 $u \in C^1(\overline{Q}) \cap C^2(Q)$ 为以下定解问题(1)的解

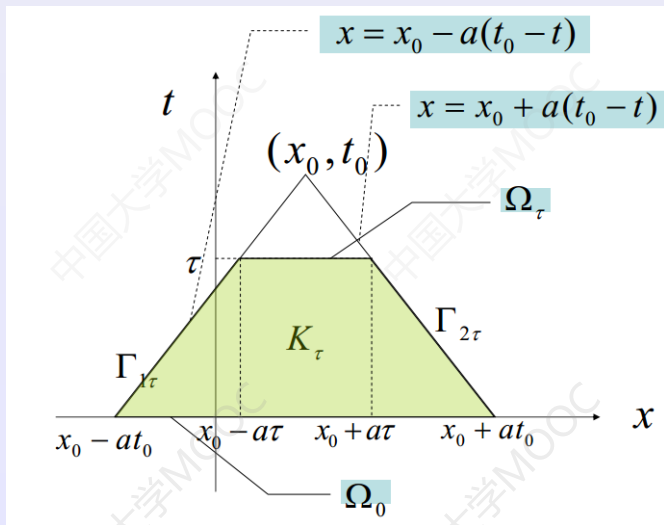
$$\begin{cases} \square u = \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t), & x \in \mathbb{R}, t > 0 \\ u(x, 0) = \varphi(x), & x \in \mathbb{R}, \\ u_t(x, 0) = \psi(x), & x \in \mathbb{R}. \end{cases} \quad (1)$$

则对于任意的 (x_0, t_0) , $t_0 > 0$, 有以下估计:

$$\int_{\Omega_\tau} [u_t^2(x, \tau) + a^2 u_x^2(x, \tau)] dx \leq M \left[\int_{\Omega_0} [\psi^2(x) + a^2 \varphi_x^2(x)] dx + \int \int_{K_\tau} f^2(x, t) dx dt \right],$$

$$\int \int_{K_\tau} [u_t^2(x, t) + a^2 u_x^2(x, t)] dx dt \leq M \left[\int_{\Omega_0} [\psi^2(x) + a^2 \varphi_x^2(x)] dx + \int \int_{K_\tau} f^2(x, t) dx dt \right],$$

其中 $0 \leq \tau \leq t_0$, $M = \exp(t_0)$.



证明: **Step 1.** 方程两边同乘以 $\frac{\partial u}{\partial t}$ 并且在 K_τ 上积分得:

$$\int \int_{K_\tau} \left(\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} \right) \frac{\partial u}{\partial t} dx dt = \int \int_{K_\tau} \frac{\partial u}{\partial t} f(x, t) dx dt$$

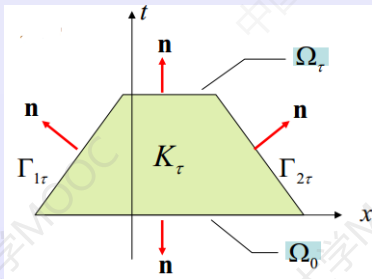
Step 2. 应用分部积分公式计算积分

$$\int \int_{K_\tau} \frac{\partial^2 u}{\partial t^2} \frac{\partial u}{\partial t} dx dt = \frac{1}{2} \int \int_{K_\tau} \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial t} \right)^2 dx dt = \frac{1}{2} \oint_{\partial K_\tau} \left(\frac{\partial u}{\partial t} \right)^2 n_t ds$$

这里: $\mathbf{n}$ 是 ∂K_τ 的单位外法向量, $\mathbf{n} = (n_x, n_t)$. $\partial K_\tau = \Omega_\tau \cup \Omega_0 \cup \Gamma_{2\tau} \cup \Gamma_{1\tau}$,
 Ω_τ 上 $\mathbf{n} = (0, 1)$, Ω_0 上 $\mathbf{n} = (0, -1)$, $\Gamma_{1\tau}$ 上 $\mathbf{n} = \frac{(-1, a)}{\sqrt{1+a^2}}$, $\Gamma_{1\tau}$ 上 $\mathbf{n} = \frac{(1, a)}{\sqrt{1+a^2}}$.

所以

$$\begin{aligned} \int \int_{K_\tau} \frac{\partial^2 u}{\partial t^2} \frac{\partial u}{\partial t} dx dt &= \frac{1}{2} \oint_{\partial K_\tau} \left(\frac{\partial u}{\partial t} \right)^2 n_t ds \\ &= \frac{1}{2} \int_{\Omega_\tau} \left(\frac{\partial u}{\partial t} \right)^2 ds - \frac{1}{2} \int_{\Omega_0} \left(\frac{\partial u}{\partial t} \right)^2 ds + \frac{1}{2} \int_{\Gamma_{1\tau}} \left(\frac{\partial u}{\partial t} \right)^2 n_t ds + \frac{1}{2} \int_{\Gamma_{2\tau}} \left(\frac{\partial u}{\partial t} \right)^2 n_t ds. \end{aligned}$$



类似地, 考虑 $\frac{\partial}{\partial x}(\frac{\partial u}{\partial x} \frac{\partial u}{\partial t})$, 可得

$$\int \int_{K_\tau} \frac{\partial^2 u}{\partial x^2} \frac{\partial u}{\partial t} dx dt = - \int \int_{K_\tau} \frac{\partial^2 u}{\partial x \partial t} \frac{\partial u}{\partial t} dx dt + \oint_{\partial K_\tau} \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} n_x ds.$$

同前面一样, 考虑 $\frac{1}{2} \frac{\partial}{\partial x} (\frac{\partial u}{\partial t})^2$, 可得

$$\begin{aligned} - \int \int_{K_\tau} \frac{\partial^2 u}{\partial x \partial t} \frac{\partial u}{\partial t} dx dt &= - \frac{1}{2} \int_{\Omega_\tau} \left(\frac{\partial u}{\partial t} \right)^2 ds + \frac{1}{2} \int_{\Omega_0} \left(\frac{\partial u}{\partial t} \right)^2 ds \\ &\quad - \frac{1}{2} \int_{\Gamma_{1\tau}} \left(\frac{\partial u}{\partial t} \right)^2 n_t ds - \frac{1}{2} \int_{\Gamma_{2\tau}} \left(\frac{\partial u}{\partial t} \right)^2 n_t ds. \end{aligned}$$

而

$$\oint_{\partial K_\tau} \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} n_x ds = \int_{\Gamma_{1\tau}} \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} n_x ds + \int_{\Gamma_{2\tau}} \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} n_x ds$$

所以

$$\begin{aligned} \int \int_{K_\tau} \frac{\partial^2 u}{\partial x^2} \frac{\partial u}{\partial t} dx dt &= -\frac{1}{2} \int_{\Omega_\tau} \left(\frac{\partial u}{\partial t} \right)^2 ds + \frac{1}{2} \int_{\Omega_0} \left(\frac{\partial u}{\partial t} \right)^2 ds \\ &\quad - \frac{1}{2} \int_{\Gamma_{1\tau} \cup \Gamma_{2\tau}} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 n_t - 2 \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} n_x \right\} ds. \end{aligned}$$

于是

$$\begin{aligned} & \int \int_{K_\tau} \frac{\partial^2 u}{\partial t^2} \frac{\partial u}{\partial t} dx dt - a^2 \int \int_{K_\tau} \frac{\partial^2 u}{\partial x \partial t} \frac{\partial u}{\partial t} dx dt \\ &= \frac{1}{2} \int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} ds - \frac{1}{2} \int_{\Omega_0} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} ds \\ & \quad + \frac{1}{2} \int_{\Gamma_{1\tau} \cup \Gamma_{2\tau}} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 n_t + a^2 \left(\frac{\partial u}{\partial x} \right)^2 n_t - 2a^2 \frac{\partial u}{\partial x} \frac{\partial u}{\partial t} n_x \right\} ds. \end{aligned}$$

注意到在 $\Gamma_{1\tau} \cup \Gamma_{2\tau}$ 上 $n_t = a|n_x| > 0$, 所以上式最后一项为

$$\frac{a|n_x|}{2} \int_{K_{1\tau} \cup K_{2\tau}} \left(\frac{\partial u}{\partial t} \pm \frac{\partial u}{\partial x} \right)^2 ds > 0,$$

得到

$$\begin{aligned}& \int \int_{K_\tau} \frac{\partial^2 u}{\partial t^2} \frac{\partial u}{\partial t} dx dt - a^2 \int \int_{K_\tau} \frac{\partial^2 u}{\partial x \partial t} \frac{\partial u}{\partial t} dx dt \\& \geq \frac{1}{2} \int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} ds - \frac{1}{2} \int_{\Omega_0} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} ds \\& = \frac{1}{2} \int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx - \frac{1}{2} \int_{\Omega_0} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx \\& = \frac{1}{2} \int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx - \frac{1}{2} \int_{\Omega_0} \{ \psi^2(x) + a^2 \varphi^2(x) \} dx.\end{aligned}$$

代入前面的等式，得

$$\begin{aligned} & \int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx \\ & \leq \int_{\Omega_0} [\psi^2(x) + a^2 \varphi^2(x)] dx + \int \int_{K_\tau} 2 \frac{\partial u}{\partial t} f(x, t) dx dt \\ & \leq \int_{\Omega_0} [\psi^2(x) + a^2 \varphi^2(x)] dx + \int \int_{K_\tau} \left(\left(\frac{\partial u}{\partial t} \right)^2 + f^2(x, t) \right) dx dt. \end{aligned}$$

Step 3. 应用Gronwall不等式

$$\begin{aligned}& \int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx \\& \leq \int_{\Omega_0} [\psi^2(x) + a^2 \varphi^2(x)] dx + \int \int_{K_\tau} f^2(x, t) dx dt + \int \int_{K_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx dt \\& = F(\tau) + G(\tau),\end{aligned}$$

其中已知值

$$F(\tau) = \int_{\Omega_0} [\psi^2(x) + a^2 \varphi^2(x)] dx + \int \int_{K_\tau} f^2(x, t) dx dt,$$

未知能量:

$$\begin{aligned} G(\tau) &= \int \int_{K_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx dt \\ &= \int_0^\tau \left\{ \int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx \right\} dt. \end{aligned}$$

因此

$$G'(\tau) = \int \int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} \Big|_{t=\tau} dx.$$

从而不等式可以表为

$$G'(\tau) \leq F(\tau) + G(\tau).$$

这是著名的**Gronwall不等式**: 由Gronwall不等式解法, 两边同乘以 $e^{-\tau}$ 得:

$$\frac{d}{d\tau} [e^{-\tau} G(\tau)] \leq e^{-\tau} F(\tau).$$

由定义知道 $G(0) = 0$, 在 $[0, \tau]$ 积分得

$$e^{-\tau} G(\tau) \leq \int_0^{\tau} e^{-x} F(x) dx.$$

解得

$$\begin{aligned} G(\tau) &\leq \int_0^{\tau} e^{\tau-x} F(x) dx \leq \int_0^{\tau} e^{\tau-x} F(\tau) dx \\ &\leq F(\tau) \int_0^{\tau} e^{\tau-x} dx \leq F(\tau)(e^{\tau} - 1). \end{aligned}$$

从而

$$\begin{aligned} G(\tau) &\leq F(\tau)(e^\tau - 1) \\ &\leq e^\tau F(\tau) \leq e^{\tau_0} F(\tau). \end{aligned}$$

所以

$$\begin{aligned} &\int_{\Omega_\tau} \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + a^2 \left(\frac{\partial u}{\partial x} \right)^2 \right\} dx \\ &= G'(\tau) \leq F(\tau) + G(\tau) \leq F(\tau) + F(\tau)(e^\tau - 1) \\ &\leq e^\tau F(\tau) \leq e^{\tau_0} F(\tau). \end{aligned}$$

定理得证.

能量估计方法总结:

第一步: 选取未知函数或其偏导数去乘方程, 并在所考虑的区域上积分。

第二步: 计算各积分, 用有关不等式放大或缩小相应积分。

第三步: (必要的时候) 利用Grönwall不等式, 估计积分。

Grönwall不等式: 若 $G'(\tau) \leq F(\tau) + G(\tau), \forall \tau \in [0, T]$;

则 $G(\tau) \leq e^\tau \int_0^\tau e^{-x} F(x) dx, \forall \tau \in [0, T]$

同学们，再见！